# Predictive model of the spread of the spotted lanternfly in the continental United States using machine learning

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Background: (Day, 2018)

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# The "What"

- Spotted lanternflies *expected to* limit production of food, timber, paper, and ornamental plants if untreated (Grove & Moltz, 2019)
- Feeding may increase shortages in plant-based resources
- Infestations may increase fungal infections in leafy plants (Cornell Integrated Pest Management, 2019)
- U.S. damage survey estimates hundreds of millions USD lost in crop damage and removal costs per year so far (FarmProgress, 2020)

(Gardner, 2018)

**More: https://github.com/Resplendent-Keklak/slf-invasion-modeling/**

# Goals

Bridge the gap between basic spotted lanternfly invasion awareness and advanced invasion forecasting using ML

1. Map historical U.S. invasion in more detail
2. Make an accurate model with a few predictors
3. Predict invasive tendencies for 2025 and 2026
4. Educate people in high-risk areas


More: https://github.com/Resplendent-Keklak/slf-invasion-modeling/

# Background

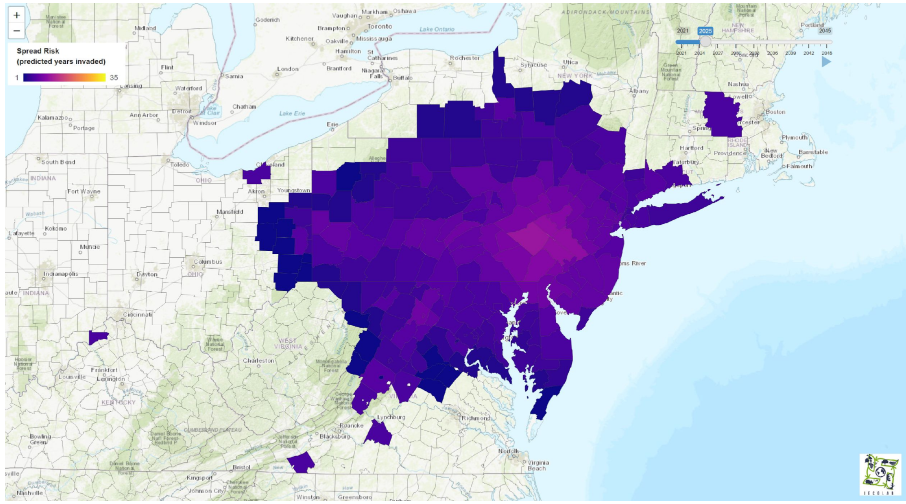
- Previous introductions to Korea saw significant risk for crop damage (Kim et al., 2021)

- Accidental introduction to Berks County, PA in 2014 by one Chinese population (Ladin et al., 2023)

- Population now much greater and spread out to almost all of the northeastern U.S.

- Data is available on the "expected greatest predictors" of invasive propagation

(Barringer, 2014)

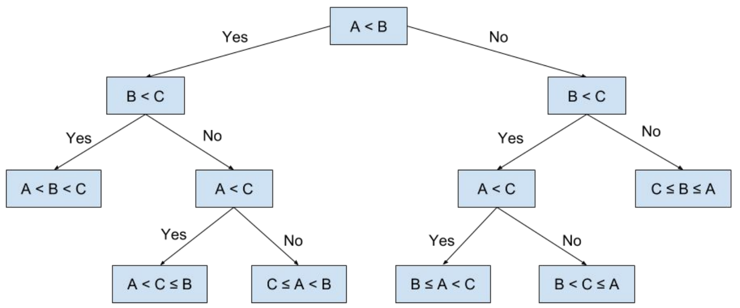


(Huron & Helmus, 2021)

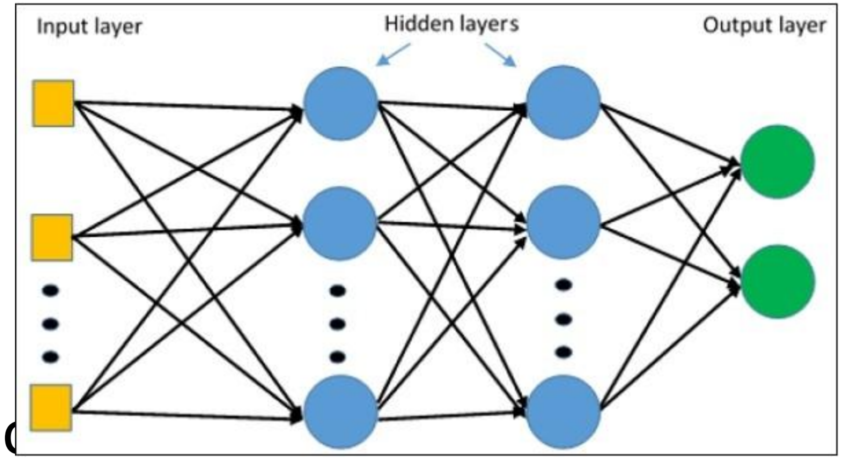
More: https://github.com/Resplendent-Keklak/slf-invasion-modeling/

# Research Approach and Methods

- Train decision trees, neural networks on each county's:
  - Estimated spotted lanternfly populations
  - Climate (temperature, humidity, sunlight, etc.)
  - Host plant abundance
  - Abundance of predators (bats, birds, insects, spiders, etc.)
  - Vehicle traffic
  - Geological features
  - Expected risk level
- Models developed using Python and existing libraries
- Trial and error: used combinations of predictors, tweak hyperparameters (features directly affecting model witho affecting predictors




(Niculaescu, 2018)




(iSolution, 2025)

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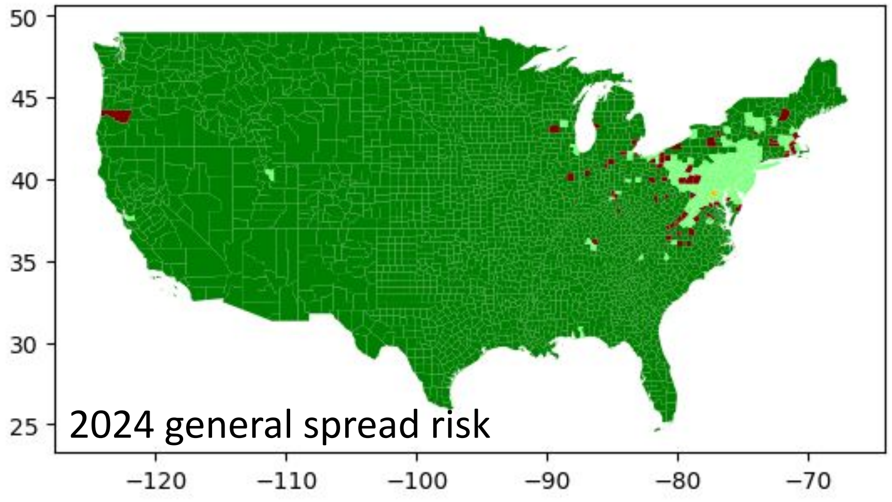
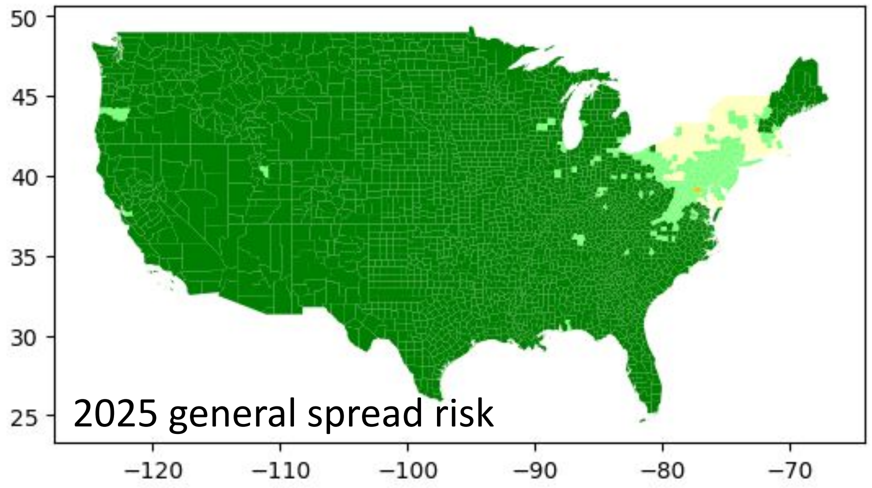
# Results (1/2)

- Proliferation risk levels in a county tend to decrease after the first year of infestation

- No host plant is safe from spotted lanternfly-induced sooty mold infection

- Couldn’t specify if the risk for future dispersal mainly originates from the interior of existing infestation zones




More: https://github.com/Resplendent-Keklak/slf-invasion-modeling/

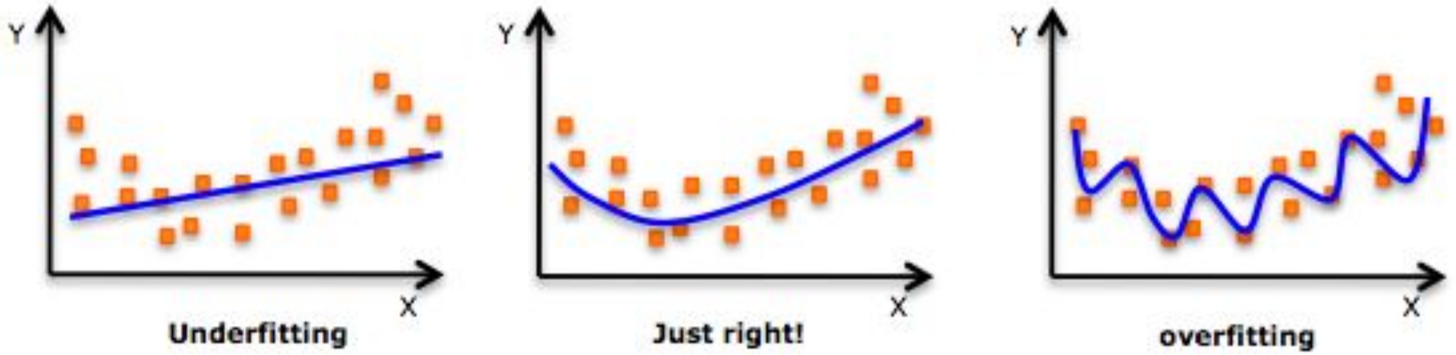
# Results (2/2)

(ISAAA, 2025)

(Schaefer, 2020)

- High risk to some purposed-grown of plants correlated with risk to plants grown for other purposes

- Very high model accuracy in cross-validation suggests current models could "overfit" and be less general

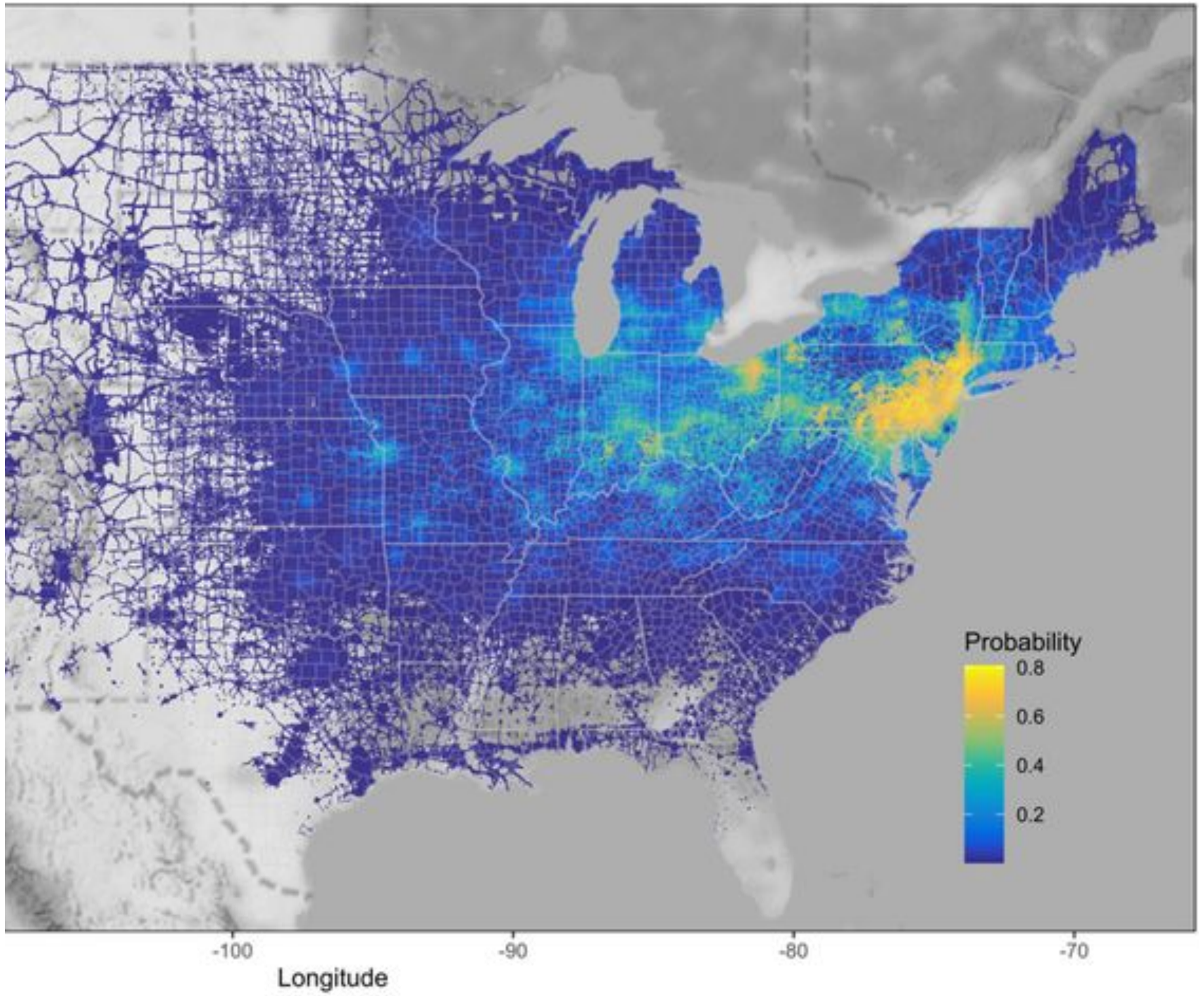



(Kumar, 2017)

**More: https://github.com/Resplendent-Keklak/slf-invasion-modeling/**

# Innovation & Discussion

- Using ML to estimate species distribution and propagation is new
- Deterministic ML sometimes performs better than stochastic ML or math models
- Supervised ML is often the easier path to this task than other simulations




(Ladin et al., 2023)

More: https://github.com/Resplendent-Keklak/slf-invasion-modeling/

# Future Work



- More public data should be available in CSV and SHP formats
- Weather data should be preserved and remain public to help w training "non-gen" AI
- Future work should involve:
  - comparing my current models to models with different predictors
  - checking how well the models predicted risk
  - comparing my ML approach to supervised regression models, stocha models
  - collecting more spotted lanternfly abundance data
  - surveying actual plant damage
  - developing and sending annual crop risk fact sheets to farmers

(Hamann, 2017)

**More: https://github.com/Resplendent-Keklak/slf-invasion-modeling/**

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More: https://github.com/Resplendent-Keklak/slf-invasion-modeling/

# Acknowledgements: Data & Software

- Publicly-accessible data providers:
  - iNaturalist and its contributors
  - U.S. Department of Agriculture (USDA)
  - U.S. Forest Service
  - National Oceanic and Atmospheric Administration (NOAA)
  - U.S. Department of Transportation (USDOT)
  - Other sources cited in repository
- Data analysis tools:
  - R and Rstudio
  - Python and Jupyter Notebook
  - Minitab
- Full data source and Python library usage list is in README.md of GitHub repository

**More: https://github.com/Resplendent-Keklak/slf-invasion-modeling/**

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**Full citation list at: https://github.com/Resplendent-Keklak/slf-invasion-modeling/**

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